LESSON 8 • TERM 4 • LEVEL 1

Graph Detectives

Read the key before you read the graph

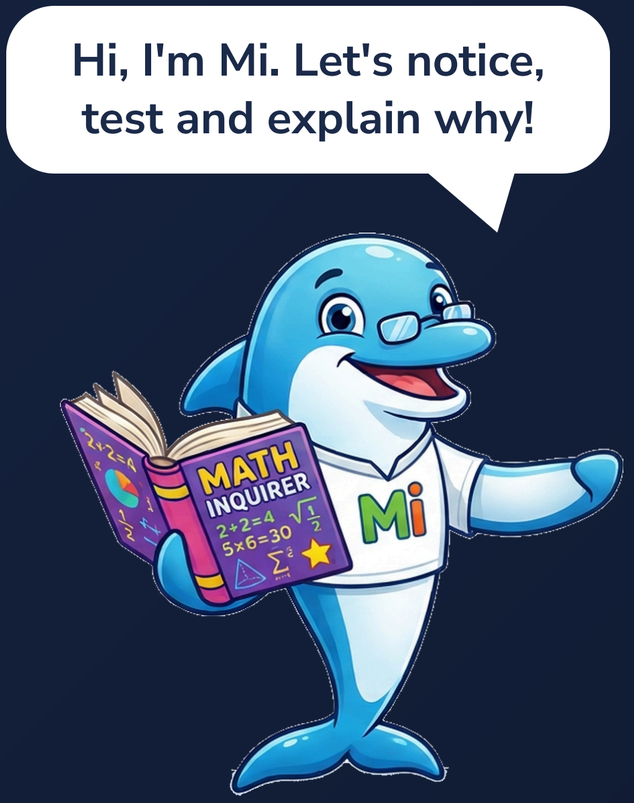
TODAY WE WILL

Read the key first.

Find a missing bar.

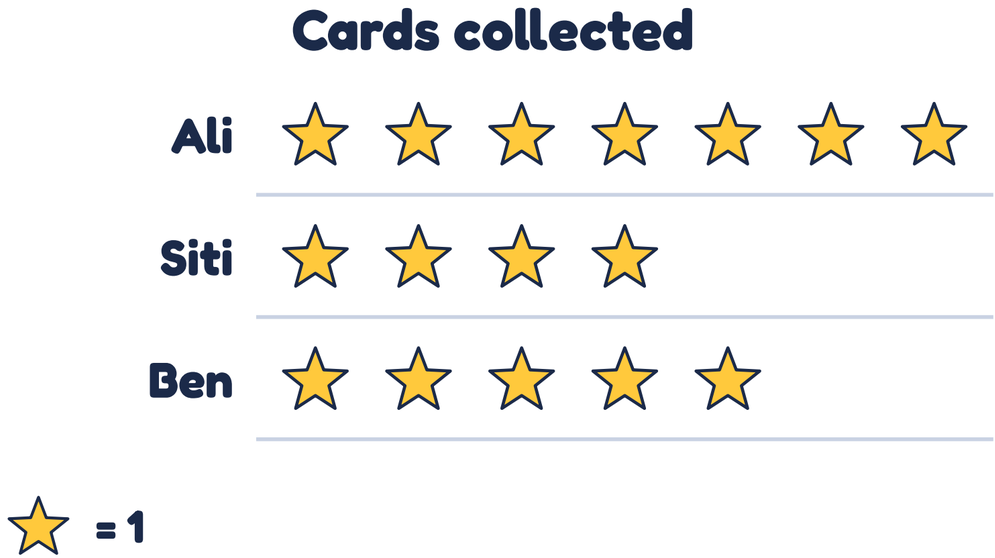
Ask what a graph cannot tell us.





Warm-up: read a graph

2–8 MIN



Each picture is 1 card. Who has the most cards? How many more cards does Ali have than Ben?

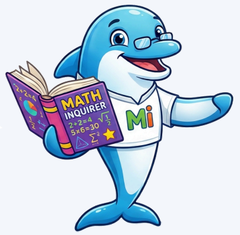
Count in 2s and 3s

2–8 MIN





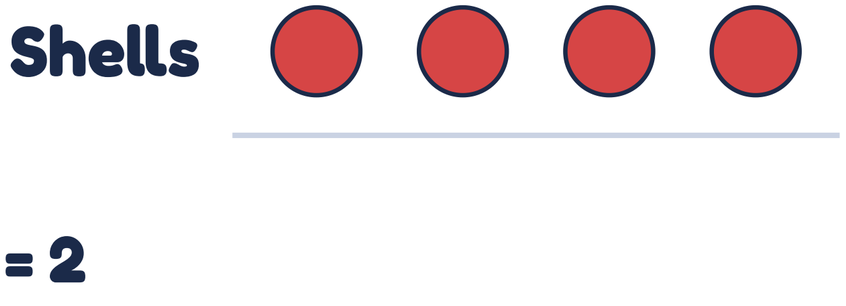
Count in 2s. Then count in 3s. Say the missing numbers.



Take your time!

Pictures that stand for 2

2–8 MIN



Each picture stands for 2. Count in 2s along the row. How many shells are there?

Sneak-peek puzzles

8–12 MIN

Ideas we have not learnt yet. A guess is a great start. Nobody is marked.

Stickers



PUZZLE 1: anyone can try

How many pictures are in Ravi's row?

Write the number.

PUZZLE 2: think

Each picture stands for 2 stickers.

How many stickers does Ravi have?

I know because ___.

Explore: make pairs

12–23 MIN

1. Draw 8 dots in a row on your paper.
2. Draw a loop around every 2 dots.
3. Each loop is 1 picture. How many pictures?

Now try 12 dots. How many pictures?

4 pictures and 6 pictures

12–23 MIN

8 dots make 4 pictures. 12 dots make 6 pictures. Each picture stands for 2 dots.

Explore: two graphs

12–23 MIN

Graph A

Pens 

 = 1

Graph B

Rulers 

 = 2

Both rows have 4 pictures. Which row shows more things? How do you know?

Read the key first

12–23 MIN

Graph A

Pens ● ● ● ●

● = 1

Graph B

Rulers ● ● ● ●

● = 2

Graph B shows more. Graph A shows 4 pens. Graph B shows 4 pictures of 2: 8 rulers.

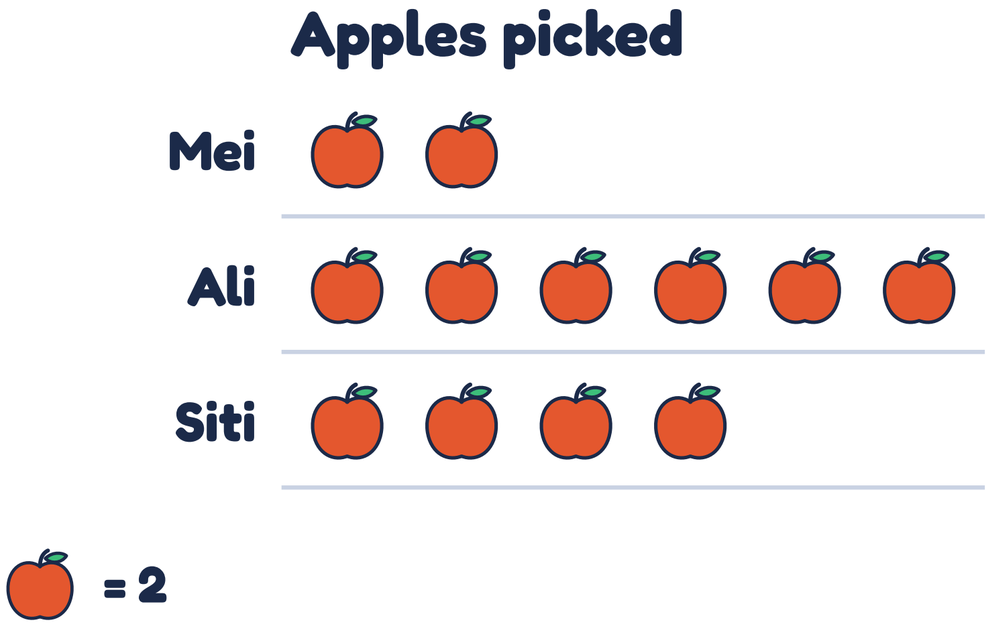
What did we notice?

12–23 MIN

- What does the key tell us?
- Is the longest row always the most things?

Read the key first

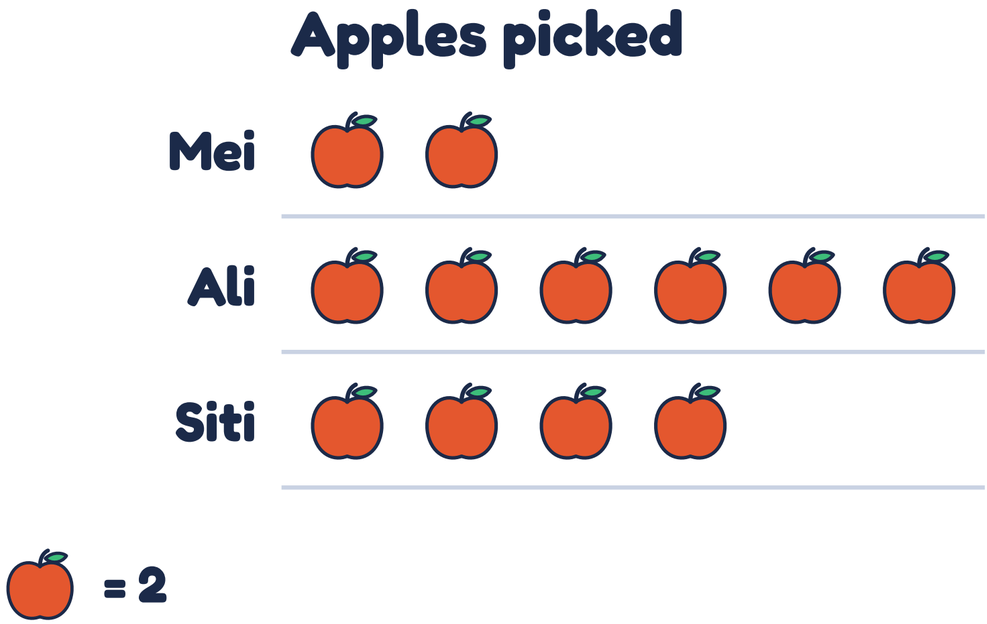
23–34 MIN



Read the key first. How many apples did Ali pick? How many more apples did Ali pick than Mei?

Ali picked 12 apples

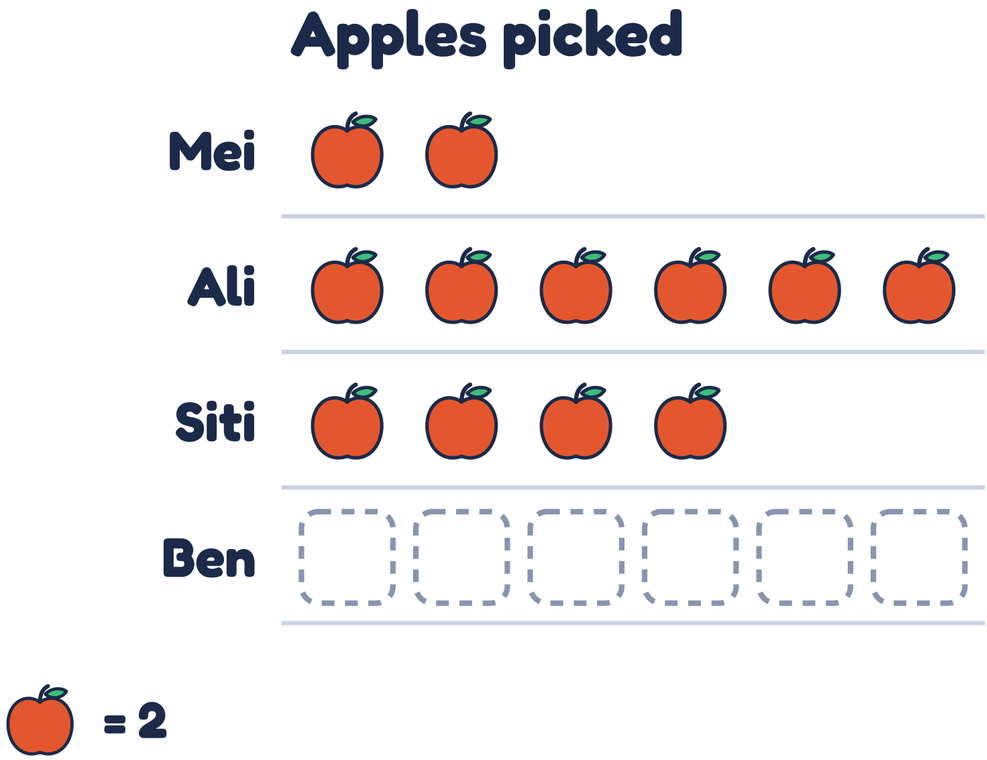
23–34 MIN



Ali picked 12 apples: 2, 4, 6, 8, 10, 12. Ali picked 8 more apples than Mei: 4 more pictures of 2.

Draw with a key

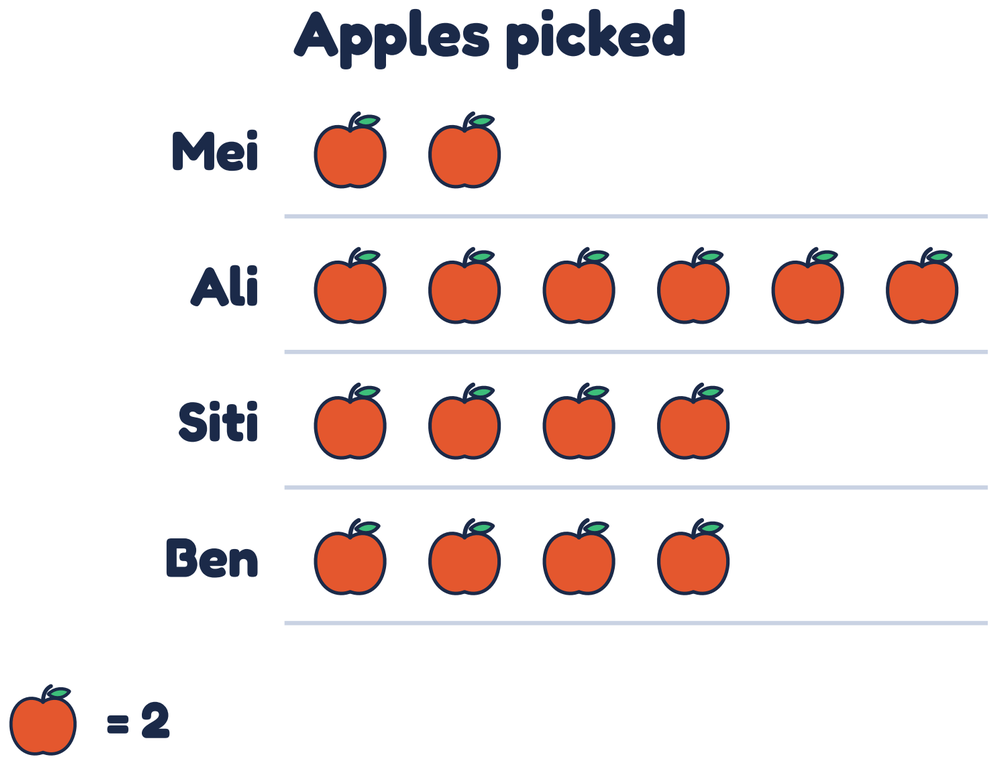
23–34 MIN



Ben picked 8 apples. Each picture stands for 2. Draw Ben's row.

4 pictures for 8 apples

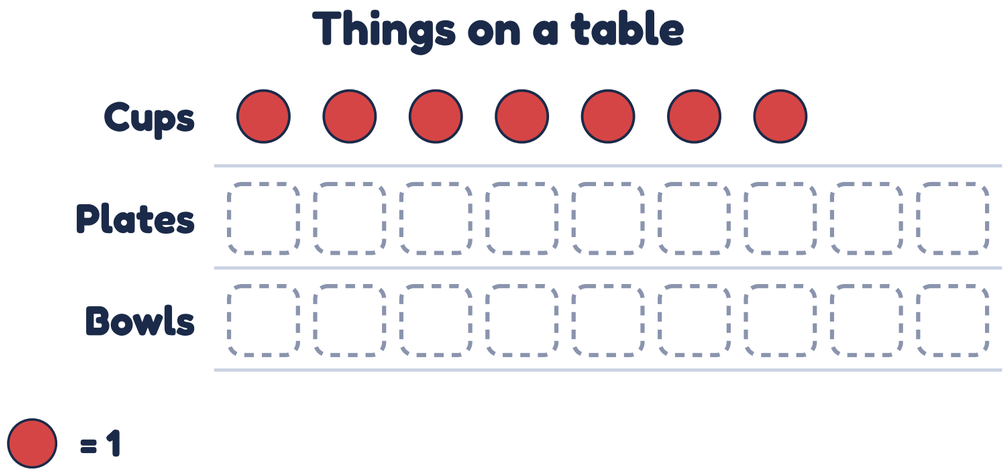
23–34 MIN



8 apples: 2, 4, 6, 8. That is 4 pictures.

The missing bar

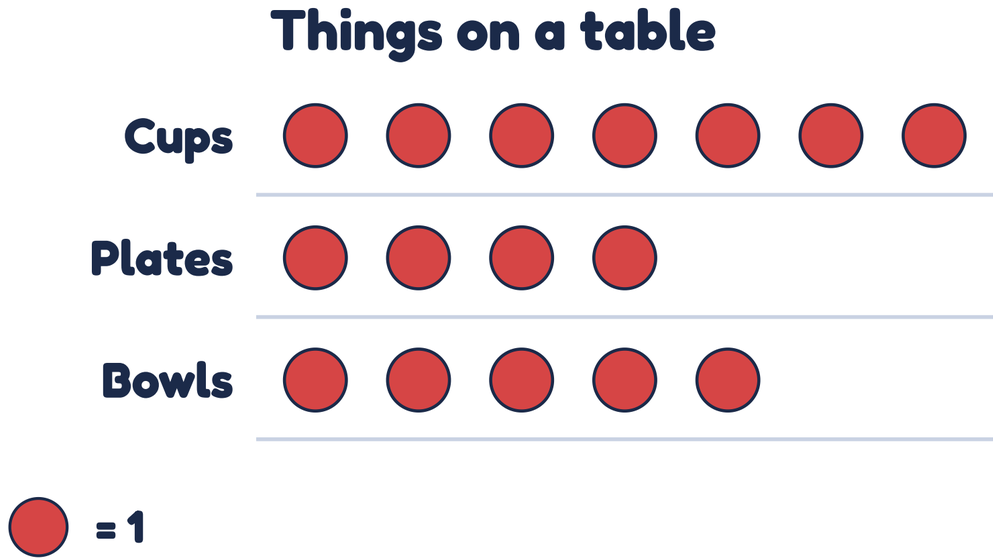
23–34 MIN



Each picture stands for 1. There are 16 things in all. There are 3 fewer plates than cups. Draw the plates and the bowls.

4 plates and 5 bowls

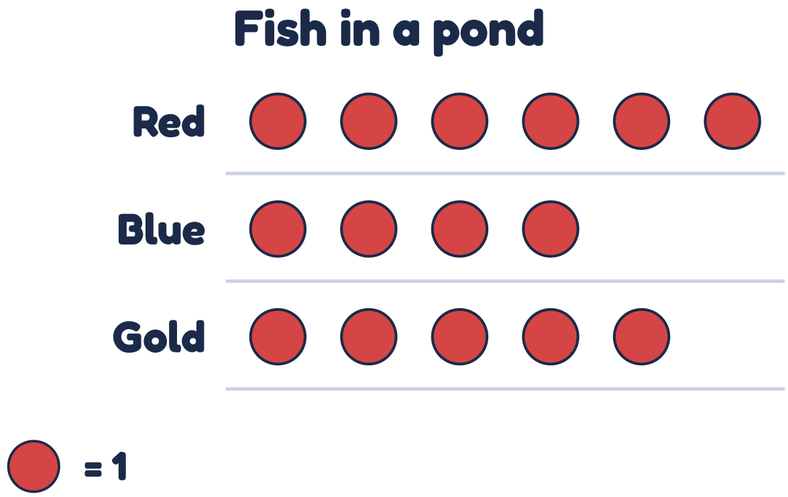
23–34 MIN



7 take away 3 is 4 plates. 7 + 4 = 11. 16 take away 11 is 5 bowls. Check: 7 + 4 + 5 = 16.

What can a graph not tell us?

23–34 MIN



1. How many gold fish are there?
2. Which fish is the happiest?
3. How many more red fish than blue fish?

Which question can the graph not answer? How do you know?

Question 2 cannot be answered

23–34 MIN

Question 2. The graph shows how many fish of each colour there are. It does not show feelings.

34–38 MIN

Count and move break

Stand up. Count in 2s and jump each time.

Now count in 3s and clap each time.

Sip some water.

Workbook page 1: Start here

38–70 MIN

Questions 1 to 3. Take your time.

S1 Study the graph. Each picture stands for 1 book. Answer questions 1 to 3.



S2 Look at the graph in question 1.

S3 Look at the graph in question 1.

Check page 1

38–70 MIN

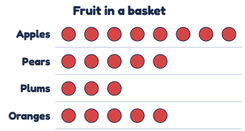
S1	Mei; Siti
S2	5 more books; 25 books in all
S3	True, True, False, False

Workbook page 2: Start here

38–70 MIN

Questions 4 to 6. Take your time.

S4 Study the graph. Each picture stands for 1 fruit. Answer questions 4 to 6.



S5 Look at the graph in question 4.

S6 Look at the graph in question 4.

38–70 MIN

Check page 2

S4	4 kinds; Oranges
S5	2 fewer plums than pears; 11 fruits
S6	21 fruits in all; Apples

38–70 MIN

Choose your next page

Page	What it is
Think harder (page 3)	A graph from clues, a key of 2, why the longest row is not always the most
Go further (page 4)	Missing Bar, a graph from clues, which question it cannot answer
Investigation (page 5)	Look, count and ask
Check	Finish and check your own work

Everyone may choose. Not finished? Pages 3 to 5 are for home too.

Talk and explain

70–76 MIN

A CHILD IN ANOTHER CLASS SAYS

There are 21 books in all.

The first bar has 6 books and the second has 9 books.

So the third bar has 8 books.

Do you agree? How could we check?

I think … because …

Back to the sneak-peek

70–76 MIN

Puzzle 1: 3 pictures. Puzzle 2: 6 stickers. Each picture stands for 2, so count in 2s: 2, 4, 6.

Exit ticket

76–82 MIN

On your own. No looking back.

E1 **Look at the graph. Each picture stands for 1 sticker.**

The fewest stickers belong to (circle one).



E2 **Look at the graph in E1.**

The children have ☐ stickers altogether.

E3 **Look at the graph in E1. Mei has 3 more stickers than Ali.**

Circle True or False.

Check the exit ticket

76–82 MIN

E1	Siti
E2	16 stickers
E3	True

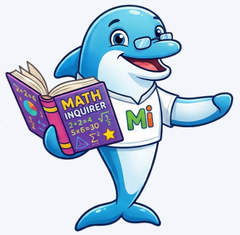
Look-back

82–85 MIN

Three minutes. There are no right answers. You can pass. Nobody reads it out.

Something I did well	write or draw one thing
The hardest part was …	☐ reading the key ☐ counting in 2s ☐ missing bars ☐ something else
What helped me	☐ reading the key first ☐ counting in 2s ☐ adding to check ☐ I was not stuck
Next time I will try	☐ reading the key ☐ missing bars ☐ questions a graph cannot answer ☐ asking a question
	☐ I am still stuck (my grown-up will photograph this page for my tutor) ☐ I would like to pass

Take your time. You can pass.



85–90 MIN

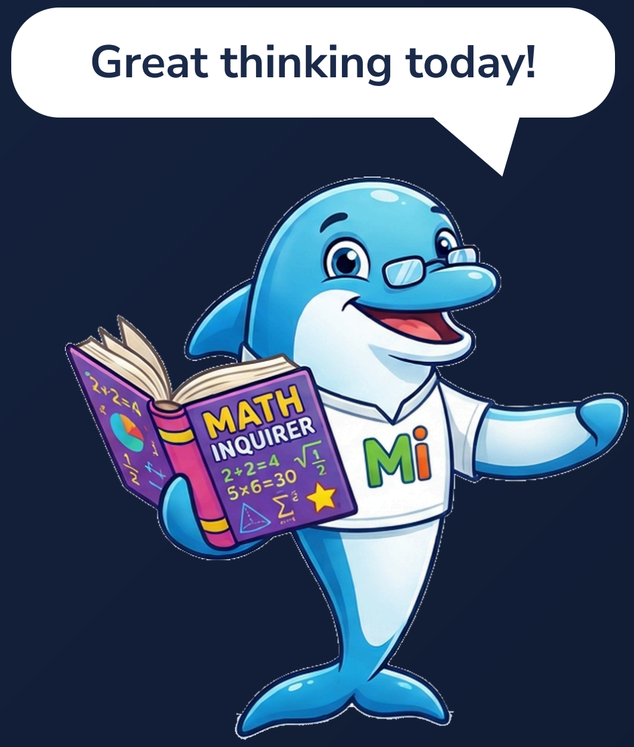
Well done, inquirers!

Show a grown-up a graph. Say what one picture stands for.

NEXT LESSON

The Problem-Solving Toolbox

Have ready: the tool-and-problem cards page from your book, with the 15 cards cut out by a grown-up.



Go further: each picture is 3

OPTIONAL

Mi's stamps

Stamps 

 = 3

Each picture stands for 3 stamps. How many stamps are in the row?

OPTIONAL

If we have time: pick one

Activity	What we do
Is it always true?	When every picture stands for 2, is the longest row always the most things?
Make a puzzle	Draw a graph with a key of 2 and hide one bar. Tell your grown-up the total.
Teach it back	Tell a toy why we read the key first.

Optional. Everyone may join.